

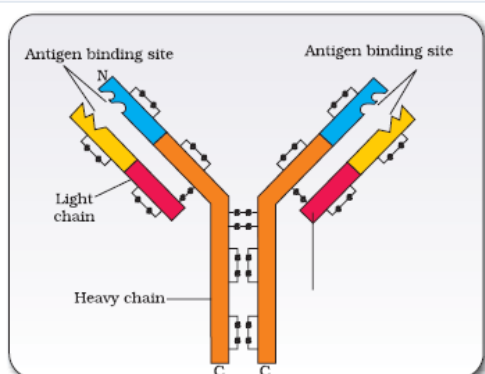
PU-II BIOLOGY · CHAPTER 7: IMPORTANT DIAGRAMS & SCHEMATIC BLUEPRINT

★ 100% GUARANTEED BOARD DIAGRAMS

DIAGRAM 1: STRUCTURE OF AN ANTIBODY MOLECULE (H_2L_2)

3 / 5 Marks

100% Expected



BOARD EXAM MARKING RUBRIC (3–5 M)

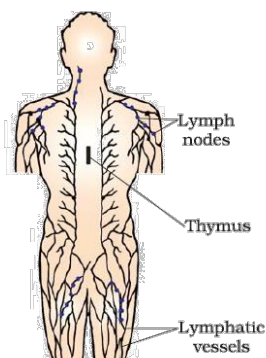
- **Y-Shape Symmetry & 4 Chains:** 2 Long Heavy (H) chains + 2 Short Light (L) chains formulated as H_2L_2 (1 Mark).
- **Disulfide Linkages (-S-S-):** Hinge inter-heavy bonds and inter-chain heavy-light bonds (1 Mark).
- **Variable (V) & Constant (C) Domains:**
 - Variable region (V_H , V_L) at amino tips forming **paratope** (Antigen-binding site).
 - Constant region (C_H , C_L) at carboxy ends (1 Mark).
- **Termini Labeling:** N-terminal (amino tips) and C-terminal (carboxy stems) correctly indicated (1 Mark).

Quick Tip: Board examiners verify: (1) 4 distinct chains, (2) -S-S- bonds between chains, (3) Variable tips labeled as Antigen-binding site, (4) Heavy vs Light chains labeled.

DIAGRAM 2: DIAGRAMMATIC REPRESENTATION OF LYMPHOID ORGANS

3 Marks

High Probability



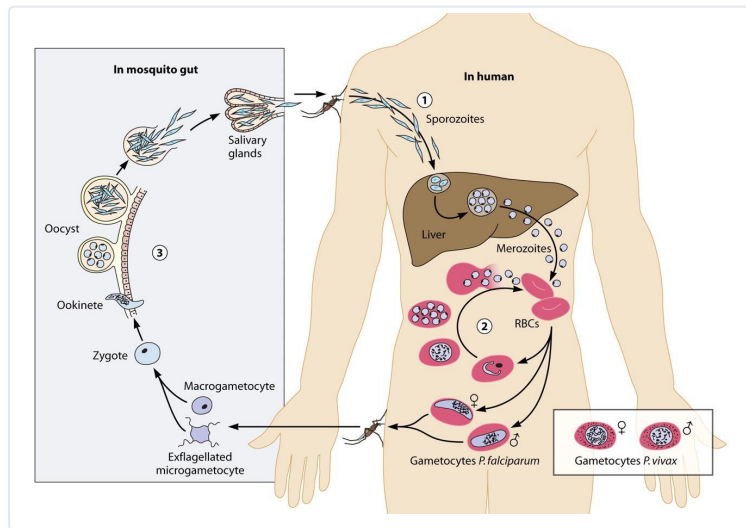
KEY ANATOMICAL LABELS & FUNCTIONS

- **Thymus:** Bilobed organ situated near the heart, behind sternum. Maturation hub for *T-lymphocytes*; progressively atrophies with age.
- **Bone Marrow:** Generative hematopoietic tissue inside long bones; produces all blood elements and site of *B-cell differentiation*.
- **Spleen:** Large bean-shaped organ on the left side of upper abdomen; phagocytoses worn-out erythrocytes and blood-borne antigens.
- **Lymph Nodes:** Nodular filters distributed along lymphatic channels; trap particulate antigens and trigger clonal lymphocyte activation.
- **MALT:** Submucosal lymphoid follicles (Peyer's patches, appendix) safeguarding mucosal tracts (50% of lymphoid tissue).

DIAGRAM 3: SCHEMATIC REPRESENTATION OF LIFE CYCLE OF *PLASMODIUM VIVAX*

5 Marks

Guaranteed 5-Marker



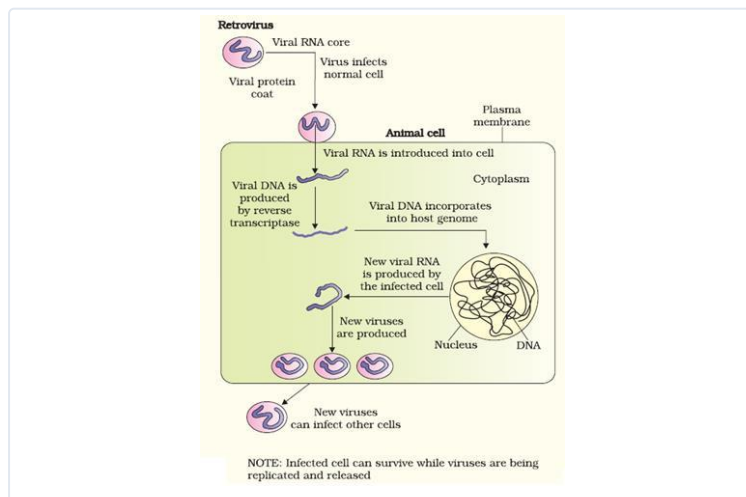
STEP-BY-STEP FLOWCHART GUIDE (5 MARKS)

- 1. Inoculation:** Needle-like infective **sporozoites** injected into human bloodstream during mosquito bite.
- 2. Hepatic Schizogony:** Sporozoites reach liver hepatocytes, divide asexually, and burst liver cells releasing **merozoites** into blood.
- 3. Erythrocytic Schizogony:** Merozoites infect RBCs; rupture releases **hemozoin** granules (causes violent chills & fever every 3–4 days).
- 4. Gametocytogenesis:** Merozoites develop into male (♂) and female (♀) **gametocytes** in human RBCs.
- 5. Syngamy in Mosquito Gut:** Female mosquito ingests gametocytes; fertilization in gut forms motile **ookinete**.
- 6. Sporogony:** Ookinete crosses gut wall → encysts as **oocyst** → sporogony forms thousands of sporozoites.
- 7. Salivary Storage:** Oocyst bursts; sporozoites lodge in mosquito **salivary glands**.

DIAGRAM 4: REPLICATION OF RETROVIRUS (HIV) INSIDE HOST CELL

5 Marks

High Frequency

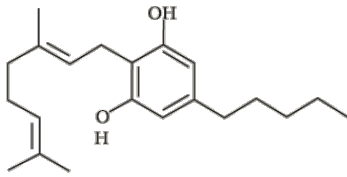


REPLICATION SEQUENCE & KEY HIGHLIGHTS

- 1. Viral Binding:** Glycoprotein spikes (gp120) bind to **CD4 surface receptors** of host macrophage or Helper T-cell.
 - 2. Uncoating:** Viral RNA core is introduced into host cytoplasm.
 - 3. Reverse Transcription:** Viral enzyme *Reverse Transcriptase* copies viral RNA into complementary viral DNA.
 - 4. Proviral Integration:** Viral DNA enters nucleus and integrates into host chromosome via *Integrase*.
 - 5. Viral Churning:** Host cell directs transcription of new viral RNA and proteins.
 - 6. Assembly & Budding:** New progeny virions assemble and bud off, infecting surrounding T_H cells.
- Crucial Note:** Macrophages survive during this cycle, functioning continuously as an "**HIV Factory**".

DIAGRAM 5: CHEMICAL STRUCTURE OF MORPHINE

2 / 3 Marks

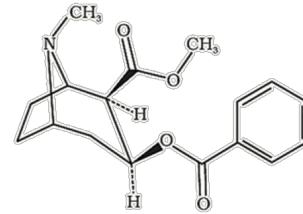


Morphine Chemical Points:

- Extracted from latex of Opium poppy (*Papaver somniferum*).
- Binds to specific opioid receptors in the **CNS and GI tract**.
- **Heroin (Smack)** is chemically **diacetylmorphine**, obtained by acetylation of morphine.

DIAGRAM 6: SKELETAL CANNABINOID MOLECULE

2 / 3 Marks



Cannabinoid Chemical Points:

- Derived from inflorescences, leaves, and resins of *Cannabis sativa*.
- Binds to **cannabinoid receptors in the brain**.
- Markedly affects the **cardiovascular system** (induces tachycardia).

DIAGRAM 7: BOTANICAL MORPHOLOGICAL SOURCES (POPPY, CANNABIS, DATURA)

2 / 3 Marks

Identification Based



Opium Poppy
(*Papaver somniferum*)



Cannabis Leaf
(*Cannabis sativa*)



Datura Branch
(*Datura stramonium*)

Board Identification Pointers:

- *Opium Poppy*: Unripe globular capsule with pink flowers; lanced to exude white milky latex rich in morphine.
- *Cannabis sativa*: Characterized by distinct serrated palmate leaflets; resin and flower tops yield marijuana, hashish, charas, and ganja.
- *Datura*: Funnel-shaped solitary actinomorphic flowers; leaves and seeds contain anticholinergic hallucinogenic tropane alkaloids.

MASTER BOARD DIAGRAM CHECKLIST & MAXIMUM MARKS WEIGHTAGE

Diagram Name	Expected Marks	Key Evaluated Features	Frequency
Antibody Molecule	3 to 5 Marks	Y-shape, H_2L_2 chains, -S-S- disulfide bonds, variable tips (antigen-binding sites), C/N terminals	Almost Every Year
Life Cycle of <i>Plasmodium</i>	5 Marks	Two-host segregation (Human vs Mosquito), hepatic schizogony, hemozoin, gut fertilization, oocyst, salivary gland sporozoites	Every Alternate Year
HIV / Retrovirus Replication	3 to 5 Marks	Attachment to CD4, reverse transcription, integration to host DNA, macrophage viral factory, budding virions	High Frequency
Lymphoid Organs Schematic	2 to 3 Marks	Location of Thymus, Bone marrow, Spleen, Lymph nodes, and MALT representation	Moderate Frequency
Morphine & Cannabinoid Structures	2 Marks	Phenanthrene core / opioid ester skeleton (morphine); Resorcinol terpene core with pentyl chain (cannabinoid)	Regular 2-Marker